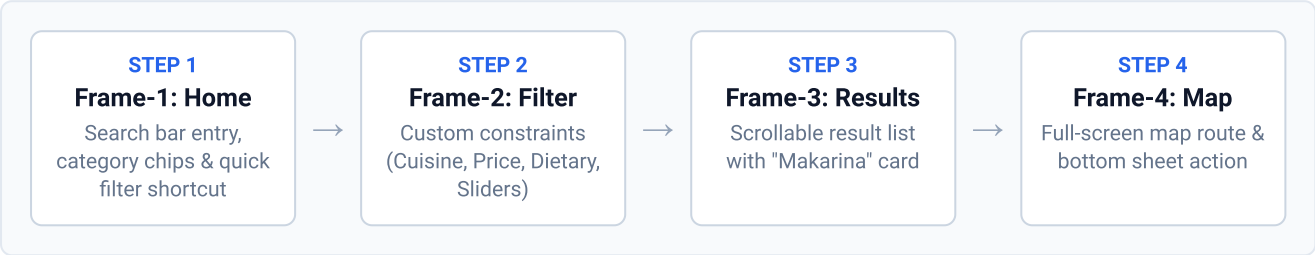
MTM4186 MOBILE APPLICATION MILESTONE • USER FLOW

User Flow Diagram & State Specifications

Application: Mobile Restaurant Search & Navigation | Scope: Core Journey & Edge Cases

1. Overview & Primary Navigation Paths

The application provides an intuitive 4-step linear flow designed to help users search, filter, evaluate, and navigate to dining venues with minimal cognitive overhead. Below is the primary path from discovery to navigation.



2. Step-by-Step Decision Matrix

Source Screen	User Interaction	System Processing / Payload	Destination
Frame-1 (Home)	Tap Filter Icon Primary	Loads current filter parameters state into memory	**Frame-2 (Filters)**
Frame-1 (Home)	Type Query + Enter	Executes direct text backend search query	**Frame-3 (Results)**
Frame-2 (Filters)	Tap "Apply" Button	Submits 18-parameter payload (Halal, Price, Radius)	**Frame-3 (Results)**
Frame-2 (Filters)	Tap "X" / Dismiss	Discards pending filter adjustments	**Frame-1 (Home)**
Frame-3 (Results)	Tap "Makarina" Card	Fetches venue details & route polyline coordinates	**Frame-4 (Map)**
Frame-4 (Map)	Tap Floating Back (<)	Dismisses map view and restores scroll position	**Frame-3 (Results)**

3. Edge Cases & Exception Handling

No Results Found: If constraints produce zero venues, Frame-3 displays a clean empty state with a "Reset Filters" action returning to default parameters.

Location Disabled: If GPS permissions are turned off, a system alert requests location access and defaults search origin to university campus coordinates.